Zomato Dataset Analysis Report

1. Introduction

Objective

The objective of this project is to analyze the Zomato restaurant dataset and extract meaningful insights related to restaurant ratings, cuisines, locations, and pricing. The analysis helps understand customer preferences and factors that influence restaurant ratings.

Dataset

The dataset was obtained from Kaggle and contains information about restaurants, including ratings, cuisines, locations, costs, and customer-related details.

2. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records.
- Handled missing values in important columns.
- Converted rating values into numeric format.
- Cleaned cost-related fields by removing commas and special characters.
- Standardized text columns for analysis.

These steps improved data quality and ensured accurate analysis.

3. Exploratory Data Analysis (EDA)

Rating Distribution

The majority of restaurants have ratings between 3.5 and 4.5, indicating generally positive customer experiences.

Cuisine Analysis

Certain cuisines appear more frequently than others, showing strong customer demand and market popularity.

Location Analysis

Restaurant concentration varies significantly across locations, with some areas acting as major dining hotspots.

Price Analysis

Restaurant pricing differs across locations and cuisine types. Cost alone does not guarantee higher customer ratings.

4. Visualizations

The following visualizations were created:

1. Restaurant Rating Distribution
2. Top Cuisines Analysis
3. Location Hotspots
4. Price vs Rating Scatter Plot
5. Correlation Heatmap
6. Cuisine WordCloud

These visualizations helped identify trends and relationships within the dataset.

5. Key Findings

- Most restaurants maintain ratings between 3.5 and 4.5.
- Popular cuisines dominate restaurant listings.
- Some locations have significantly higher restaurant density than others.
- The relationship between price and rating is relatively weak.
- Customer preferences vary according to cuisine and location.

6. Recommendations for Alfido Tech Platform

1. Promote Highly Rated Restaurants

Feature top-rated restaurants to improve user trust and engagement.

2. Build Cuisine-Based Recommendations

Suggest restaurants based on user cuisine preferences.

3. Focus on High-Demand Locations

Increase partnerships and marketing efforts in restaurant hotspot areas.

4. Utilize Customer Feedback

Analyze reviews and ratings to identify customer preferences and improvement opportunities.

5. Provide Pricing Insights

Offer restaurant owners comparative pricing analytics to remain competitive.

7. Conclusion

This project successfully analyzed restaurant data from the Zomato dataset using data cleaning, exploratory data analysis, and visualization techniques. The findings provide valuable insights into customer preferences, restaurant performance, and business opportunities. The recommendations can help platforms like Alfido Tech improve user experience, restaurant partnerships, and data-driven decision-making.